Practical 10

a. Using JavaScript design, a web page that prints factorial/Fibonacci series/any given series.

index.html

```
<!DOCTYPE html>
<html>
<head>
    <title>10(a) - Factorial and Fibonacci</title>

    <style>
        body {
            font-family: Arial, sans-serif;
            background-color: #f2f2f2;
            padding: 20px;
        }

        section {
            background-color: white;
            padding: 20px;
            margin-bottom: 20px;
            border: 2px solid #333;
            border-radius: 10px;
        }

        .box {
            margin: 15px 0;
            padding: 15px;
            background-color: #eeeeee;
            border-radius: 8px;
        }

        input, button {
            padding: 8px;
            margin: 5px;
        }

        button {
            cursor: pointer;
        }

        #result {
            font-weight: bold;
            color: blue;
        }
    </style>
</head>

<body>

    <section>
        <h1>Practical 10(a)</h1>
        <h2>Factorial / Fibonacci Series / Given Series</h2>
```

```
        <div class="box">
            <h3>Factorial</h3>

            <label>Enter a number:</label>
            <input type="number" id="factNum">

            <button onclick="findFactorial()">Find Factorial</button>

            <p id="factorialResult"></p>
        </div>

        <div class="box">
            <h3>Fibonacci Series</h3>

            <label>Enter number of terms:</label>
            <input type="number" id="fibNum">

            <button onclick="fibonacci()">Generate Fibonacci</button>

            <p id="fibonacciResult"></p>
        </div>

        <div class="box">
            <h3>Any Given Series</h3>

            <p>Example: 1, 2, 3, 4, ...</p>

            <label>Enter number of terms:</label>
            <input type="number" id="seriesNum">

            <button onclick="generateSeries()">Generate Series</button>

            <p id="seriesResult"></p>
        </div>
    </section>

    <script>

        // Factorial
        function findFactorial() {

            let n = Number(document.getElementById("factNum").value);
            let fact = 1;

            if (n < 0) {
                document.getElementById("factorialResult").innerHTML =
                    "Factorial is not defined for negative numbers.";
                return;
            }

            for (let i = 1; i <= n; i++) {
                fact = fact * i;
            }

            document.getElementById("factorialResult").innerHTML =
                "Factorial of " + n + " = " + fact;
```

```
        }


        // Fibonacci Series
        function fibonacci() {

            let n = Number(document.getElementById("fibNum").value);

            let a = 0;
            let b = 1;
            let result = "";

            for (let i = 1; i <= n; i++) {

                result += a + " ";

                let next = a + b;
                a = b;
                b = next;
            }

            document.getElementById("fibonacciResult").innerHTML =
                "Fibonacci Series: " + result;
        }


        // Given Series: 1, 2, 3, 4, ...
        function generateSeries() {

            let n = Number(document.getElementById("seriesNum").value);
            let result = "";

            for (let i = 1; i <= n; i++) {
                result += i + " ";
            }

            document.getElementById("seriesResult").innerHTML =
                "Series: " + result;
        }

    </script>

</body>
</html>
```

b. Design a form and validate all the controls placed on the form using Java Script.

index.html

```
<!DOCTYPE html>
<html>
<head>
    <title>10(b) - Form Validation</title>

    <style>
        body {
            font-family: Arial, sans-serif;
            background-color: #f2f2f2;
            padding: 20px;
        }

        section {
            max-width: 600px;
            margin: auto;
            background-color: white;
            padding: 25px;
            border: 2px solid #333;
            border-radius: 10px;
        }

        .form-control {
            margin-bottom: 15px;
        }

        label {
            display: block;
            margin-bottom: 5px;
            font-weight: bold;
        }

        input, select, textarea {
            width: 95%;
            padding: 10px;
            border: 1px solid #999;
            border-radius: 5px;
        }

        button {
            padding: 10px 20px;
            background-color: green;
            color: white;
            border: none;
            border-radius: 5px;
            cursor: pointer;
        }

        #message {
            margin-top: 15px;
            font-weight: bold;
        }
    </style>
</head>
```

```html
<body>

    <section>

        <h1>Practical 10(b)</h1>
        <h2>Form Design and Validation</h2>

        <form onsubmit="return validateForm()">

            <div class="form-control">
                <label>Name:</label>
                <input type="text" id="name">
            </div>

            <div class="form-control">
                <label>Email:</label>
                <input type="text" id="email">
            </div>

            <div class="form-control">
                <label>Age:</label>
                <input type="number" id="age">
            </div>

            <div class="form-control">
                <label>Gender:</label>

                <select id="gender">
                    <option value="">Select Gender</option>
                    <option value="Male">Male</option>
                    <option value="Female">Female</option>
                    <option value="Other">Other</option>
                </select>
            </div>

            <div class="form-control">
                <label>Password:</label>
                <input type="password" id="password">
            </div>

            <div class="form-control">
                <label>Address:</label>
                <textarea id="address"></textarea>
            </div>

            <button type="submit">Submit</button>

            <div id="message"></div>

        </form>

    </section>


    <script>
```

```
function validateForm() {

    let name = document.getElementById("name").value.trim();
    let email = document.getElementById("email").value.trim();
    let age = document.getElementById("age").value;
    let gender = document.getElementById("gender").value;
    let password = document.getElementById("password").value;
    let address = document.getElementById("address").value.trim();

    let message = document.getElementById("message");


    // Name validation
    if (name === "") {
        message.innerHTML = "Please enter your name.";
        message.style.color = "red";
        return false;
    }


    // Email validation
    if (email === "" || !email.includes("@")) {
        message.innerHTML = "Please enter a valid email.";
        message.style.color = "red";
        return false;
    }


    // Age validation
    if (age === "" || age < 1 || age > 100) {
        message.innerHTML = "Please enter a valid age.";
        message.style.color = "red";
        return false;
    }


    // Gender validation
    if (gender === "") {
        message.innerHTML = "Please select your gender.";
        message.style.color = "red";
        return false;
    }


    // Password validation
    if (password.length < 6) {
        message.innerHTML =
            "Password must contain at least 6 characters.";
        message.style.color = "red";
        return false;
    }


    // Address validation
    if (address === "") {
```

```
                message.innerHTML = "Please enter your address.";
                message.style.color = "red";
                return false;
            }


            message.innerHTML = "Form submitted successfully!";
            message.style.color = "green";

            return false;
        }

    </script>

</body>
</html>
```

c. Write a JavaScript program to display all the prime numbers between 1 and 100.

index.html

```
<!DOCTYPE html>
<html>
<head>
    <title>10(c) - Prime Numbers</title>

    <style>
        body {
            font-family: Arial, sans-serif;
            background-color: #f2f2f2;
            padding: 20px;
        }

        section {
            background-color: white;
            padding: 25px;
            border: 2px solid #333;
            border-radius: 10px;
            max-width: 600px;
            margin: auto;
        }

        .box {
            background-color: #eeeeee;
            padding: 20px;
            margin-top: 20px;
        }

        button {
            padding: 10px 20px;
            cursor: pointer;
        }

        #result {
            color: blue;
            font-size: 18px;
        }
    </style>
</head>

<body>

    <section>

        <h1>Practical 10(c)</h1>
        <h2>Prime Numbers Between 1 and 100</h2>

        <div class="box">

            <button onclick="findPrimes()">
                Display Prime Numbers
            </button>

            <p id="result"></p>
```

```
        </div>

    </section>


    <script>

        function findPrimes() {

            let primes = "";

            for (let number = 2; number <= 100; number++) {

                let isPrime = true;

                for (let i = 2; i < number; i++) {

                    if (number % i === 0) {
                        isPrime = false;
                        break;
                    }
                }

                if (isPrime) {
                    primes += number + " ";
                }
            }

            document.getElementById("result").innerHTML =
                "Prime Numbers: " + primes;
        }

    </script>

</body>
</html>
```

Practical 11

a. Write a JavaScript program to accept a number from the user and display the sum of its digits.

index.html

```
<!DOCTYPE html>
<html>
<head>
    <title>11(a) - Sum of Digits</title>

    <style>
        body {
            font-family: Arial, sans-serif;
            background-color: #f2f2f2;
            padding: 20px;
        }

        section {
            max-width: 500px;
            margin: auto;
            background-color: white;
            padding: 25px;
            border: 2px solid #333;
            border-radius: 10px;
        }

        .box {
            background-color: #eeeeee;
            padding: 20px;
        }

        input, button {
            padding: 10px;
            margin: 5px;
        }

        #result {
            font-weight: bold;
            color: blue;
        }
    </style>
</head>

<body>

    <section>

        <h1>Practical 11(a)</h1>
        <h2>Sum of Digits</h2>

        <div class="box">

            <label>Enter a number:</label>
```

```
            <input type="number" id="number">

            <button onclick="sumDigits()">
                Calculate
            </button>

            <p id="result"></p>

        </div>

    </section>


    <script>

        function sumDigits() {

            let number = Number(document.getElementById("number").value);

            number = Math.abs(number);

            let sum = 0;

            while (number > 0) {

                let digit = number % 10;

                sum = sum + digit;

                number = Math.floor(number / 10);
            }

            document.getElementById("result").innerHTML =
                "Sum of digits = " + sum;
        }

    </script>

</body>
</html>
```

b. Write a program in JavaScript to accept a sentence from the user and display the number of words in it. (Do not use split() function).

index.html

```
<!DOCTYPE html>
<html>
<head>
    <title>11(b) - Word Count</title>

    <style>
        body {
            font-family: Arial, sans-serif;
            background-color: #f2f2f2;
            padding: 20px;
        }

        section {
            max-width: 700px;
            margin: auto;
            background-color: white;
            padding: 25px;
            border: 2px solid #333;
            border-radius: 10px;
        }

        .box {
            background-color: #eeeeee;
            padding: 20px;
        }

        textarea {
            width: 95%;
            height: 120px;
            padding: 10px;
        }

        button {
            padding: 10px 20px;
            margin-top: 10px;
        }

        #result {
            color: blue;
            font-weight: bold;
        }
    </style>
</head>

<body>

    <section>

        <h1>Practical 11(b)</h1>
        <h2>Count Number of Words in a Sentence</h2>

        <div class="box">
```

```
            <label>Enter a sentence:</label>

            <br><br>

            <textarea id="sentence"></textarea>

            <br>

            <button onclick="countWords()">
                Count Words
            </button>

            <p id="result"></p>

        </div>

    </section>


    <script>

        function countWords() {

            let sentence =
                document.getElementById("sentence").value.trim();

            let count = 0;
            let inWord = false;


            // Do not use split()

            for (let i = 0; i < sentence.length; i++) {

                if (sentence[i] !== " " && sentence[i] !== "\t") {

                    if (!inWord) {
                        count++;
                        inWord = true;
                    }

                } else {

                    inWord = false;
                }
            }


            document.getElementById("result").innerHTML =
                "Number of words = " + count;
        }

    </script>

</body>
```

```
</html>
```

c. Write a JavaScript program to design a simple calculator.

index.html

```
<!DOCTYPE html>
<html>
<head>
    <title>11(c) - Simple Calculator</title>

    <style>
        body {
            font-family: Arial, sans-serif;
            background-color: #f2f2f2;
            padding: 20px;
        }

        section {
            max-width: 500px;
            margin: auto;
            background-color: white;
            padding: 25px;
            border: 2px solid #333;
            border-radius: 10px;
        }

        .calculator {
            background-color: #eeeeee;
            padding: 20px;
        }

        input, select, button {
            padding: 10px;
            margin: 5px;
        }

        #result {
            color: blue;
            font-size: 20px;
            font-weight: bold;
        }
    </style>
</head>

<body>

    <section>

        <h1>Practical 11(c)</h1>
        <h2>Simple Calculator</h2>

        <div class="calculator">

            <input type="number" id="num1" placeholder="First number">

            <br>

            <select id="operator">
```

```
            <option value="+">Addition (+)</option>
            <option value="-">Subtraction (-)</option>
            <option value="*">Multiplication (*)</option>
            <option value="/">Division (/)</option>
            <option value="%">Modulus (%)</option>
        </select>

        <br>

        <input type="number" id="num2" placeholder="Second number">

        <br>

        <button onclick="calculate()">
            Calculate
        </button>

        <p id="result"></p>

    </div>

</section>


<script>

    function calculate() {

        let num1 = Number(document.getElementById("num1").value);
        let num2 = Number(document.getElementById("num2").value);

        let operator =
            document.getElementById("operator").value;

        let result;


        if (operator === "+") {
            result = num1 + num2;
        }

        else if (operator === "-") {
            result = num1 - num2;
        }

        else if (operator === "*") {
            result = num1 * num2;
        }

        else if (operator === "/") {

            if (num2 === 0) {
                document.getElementById("result").innerHTML =
                    "Cannot divide by zero.";
                return;
            }
```

```
                result = num1 / num2;
            }

            else if (operator === "%") {
                result = num1 % num2;
            }


            document.getElementById("result").innerHTML =
                "Result = " + result;
        }

    </script>

</body>
</html>
```

Practical 12

Using JavaScript

a. Design a web page demonstrating different conditional statements.

index.html

```
<!DOCTYPE html>
<html>
<head>
    <title>12(a) - Conditional Statements</title>

    <style>
        body {
            font-family: Arial, sans-serif;
            background-color: #f2f2f2;
            padding: 20px;
        }

        section {
            background-color: white;
            padding: 25px;
            border: 2px solid #333;
            border-radius: 10px;
            max-width: 700px;
            margin: auto;
        }

        .box {
            background-color: #eeeeee;
            padding: 20px;
            margin-top: 15px;
            border-radius: 8px;
        }

        input, button {
            padding: 10px;
            margin: 5px;
        }

        #result {
            color: blue;
            font-weight: bold;
        }
    </style>
</head>

<body>

<section>

    <h1>Practical 12(a)</h1>
    <h2>Demonstration of Conditional Statements</h2>

    <!-- ================ IF CONDITION ================ -->
    <div>
```

```
        <h3>1. if Condition</h3>
        <p><b>Used for:</b> Checking whether a number is positive.</p>

        <input type="number" id="num1">
        <button onclick="ifExample()">Check</button>

        <p id="result1"></p>
    </div>


    <!-- ================= IF-ELSE CONDITION ================= -->
    <div>
        <h3>2. if-else Condition</h3>
        <p><b>Used for:</b> Checking whether a number is Even or Odd.</p>

        <input type="number" id="num2">
        <button onclick="ifElseExample()">Check</button>

        <p id="result2"></p>
    </div>


    <!-- ================= ELSE-IF CONDITION ================= -->
    <div>
        <h3>3. if-else-if Condition</h3>
        <p><b>Used for:</b> Finding grade according to marks.</p>

        <input type="number" id="marks">
        <button onclick="gradeExample()">Find Grade</button>

        <p id="result3"></p>
    </div>


    <!-- ================= SWITCH CONDITION ================= -->
    <div>
        <h3>4. switch Condition</h3>
        <p><b>Used for:</b> Displaying day according to day number.</p>

        <input type="number" id="day">
        <button onclick="switchExample()">Find Day</button>

        <p id="result4"></p>
    </div>

</section>


<script>

// ---------------------------------------------------
// 1. IF CONDITION
// Used to check a single condition
// ---------------------------------------------------

function ifExample() {
```

```
    let num = Number(document.getElementById("num1").value);

    if (num > 0) {
        document.getElementById("result1").innerHTML =
            "Number is Positive";
    }
}


// ------------------------------------------------
// 2. IF-ELSE CONDITION
// Used when there are two possibilities
// ------------------------------------------------

function ifElseExample() {

    let num = Number(document.getElementById("num2").value);

    if (num % 2 == 0) {
        document.getElementById("result2").innerHTML =
            "Number is Even";
    }
    else {
        document.getElementById("result2").innerHTML =
            "Number is Odd";
    }
}


// ------------------------------------------------
// 3. IF-ELSE-IF CONDITION
// Used when there are multiple conditions
// ------------------------------------------------

function gradeExample() {

    let marks = Number(document.getElementById("marks").value);
    let grade;

    if (marks >= 90) {
        grade = "A+";
    }
    else if (marks >= 80) {
        grade = "A";
    }
    else if (marks >= 70) {
        grade = "B";
    }
    else if (marks >= 60) {
        grade = "C";
    }
    else if (marks >= 50) {
        grade = "D";
    }
    else {
```

```
        grade = "Fail";
    }

    document.getElementById("result3").innerHTML =
        "Grade = " + grade;
}


// ------------------------------------------------
// 4. SWITCH CONDITION
// Used when one value has multiple fixed choices
// ------------------------------------------------

function switchExample() {

    let day = Number(document.getElementById("day").value);
    let dayName;

    switch (day) {

        case 1:
            dayName = "Monday";
            break;

        case 2:
            dayName = "Tuesday";
            break;

        case 3:
            dayName = "Wednesday";
            break;

        case 4:
            dayName = "Thursday";
            break;

        case 5:
            dayName = "Friday";
            break;

        case 6:
            dayName = "Saturday";
            break;

        case 7:
            dayName = "Sunday";
            break;

        default:
            dayName = "Invalid Day";
    }

    document.getElementById("result4").innerHTML =
        "Day = " + dayName;
}
```

```
</script>

</body>
</html>
```

b. Design a web page demonstrating different looping statements.

index.html

```
<!DOCTYPE html>
<html>

<head>
    <title>12(b) - Looping Statements</title>

    <style>
        body {
            font-family: Arial, sans-serif;
            background-color: #f2f2f2;
            padding: 20px;
        }

        section {
            background-color: white;
            padding: 25px;
            border: 2px solid #333;
            border-radius: 10px;
            max-width: 800px;
            margin: auto;
        }

        .box {
            background-color: #eeeeee;
            padding: 20px;
            margin: 15px 0;
            border-radius: 8px;
        }

        button {
            padding: 10px 20px;
            cursor: pointer;
        }

        p {
            color: blue;
            font-weight: bold;
        }
    </style>
</head>

<body>

    <section>

        <h1>Practical 12(b)</h1>
        <h2>Demonstration of Looping Statements</h2>


        <!-- ================= FOR LOOP ================= -->

        <div>
            <h3>1. for Loop</h3>
```

```
        <p>
            <b>Used for:</b>
            Displaying numbers from 1 to 10 when the number
            of repetitions is known.
        </p>

        <button onclick="forLoop()">Run for Loop</button>

        <p id="forResult"></p>
    </div>


    <!-- ================= WHILE LOOP ================= -->

    <div>
        <h3>2. while Loop</h3>

        <p>
            <b>Used for:</b>
            Displaying numbers from 1 to 10 while a condition
            remains true.
        </p>

        <button onclick="whileLoop()">Run while Loop</button>

        <p id="whileResult"></p>
    </div>


    <!-- ================= DO-WHILE LOOP ================= -->

    <div>
        <h3>3. do-while Loop</h3>

        <p>
            <b>Used for:</b>
            Executing the statement at least once before
            checking the condition.
        </p>

        <button onclick="doWhileLoop()">Run do-while Loop</button>

        <p id="doWhileResult"></p>
    </div>


    <!-- ================= NESTED LOOP ================= -->

    <div>
        <h3>4. Nested for Loop</h3>

        <p>
            <b>Used for:</b>
            Creating patterns using one loop inside another loop.
        </p>
```

```
        <button onclick="nestedLoop()">Run Nested Loop</button>

        <p id="nestedResult"></p>
    </div>

</section>


<script>


    // ------------------------------------------------
    // 1. FOR LOOP
    // Used when number of repetitions is known
    // Example: Print numbers 1 to 10
    // ------------------------------------------------

    function forLoop() {

        let result = "";

        for (let i = 1; i <= 10; i++) {

            result += i + " ";
        }

        document.getElementById("forResult").innerHTML =
            "For Loop: " + result;
    }


    // ------------------------------------------------
    // 2. WHILE LOOP
    // Used when repetition depends on a condition
    // ------------------------------------------------

    function whileLoop() {

        let i = 1;
        let result = "";

        while (i <= 10) {

            result += i + " ";

            i++;
        }

        document.getElementById("whileResult").innerHTML =
            "While Loop: " + result;
    }


    // ------------------------------------------------
    // 3. DO-WHILE LOOP
```

```
        // Executes at least once before checking condition
        // ------------------------------------------------

        function doWhileLoop() {

            let i = 1;
            let result = "";

            do {

                result += i + " ";

                i++;

            } while (i <= 10);

            document.getElementById("doWhileResult").innerHTML =
                "Do-While Loop: " + result;
        }


        // ------------------------------------------------
        // 4. NESTED FOR LOOP
        // One loop inside another loop
        // Used to create patterns
        // ------------------------------------------------

        function nestedLoop() {

            let result = "";

            for (let i = 1; i <= 5; i++) {

                for (let j = 1; j <= i; j++) {

                    result += "* ";
                }

                result += "<br>";
            }

            document.getElementById("nestedResult").innerHTML =
                result;
        }

    </script>

</body>

</html>
```